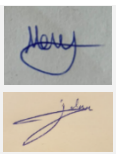
Roboteam

UGV BISON
VHCR-V01-00001
Acceptance Test Procedure

Approvals				
	Title	Name	Date	Signature
Written by	QA Engineer NPI Engineer	May Swisa Idan Kabesa	25.09.2025	
Approved By	Engineering Team Leader	Roi Akva	25.09.2025	
List of changes				
Rev	Description	Name	Role	Date
000	First release	May Swisa Idan Kabesa	QA Engineer NPI Engineer	21.09.2025
001	VHCR-V01-00001 update	Idan Kabesa	NPI Engineer	05.08.2026

1. Goal:

To verify that the Argo system's performance complies with the system specifications.

2. Relevant Documents:

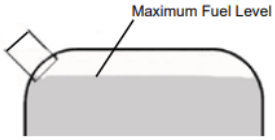
- All assembly Instructions for Argo
- All QC/ATP instructions for Argo

3. Products/Tools:

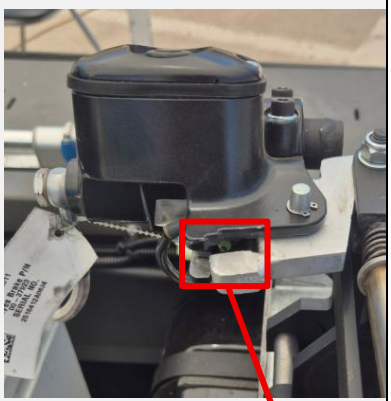
Products/Tools		Description
1	VHCR-V01-00001	BISON-UGV (FOR ARGO 8X8)
2	SET00149-A	ROBOX-GUN SYSTEM SET
3	MCH00746	ARGO ROBOT 8X8 TRACKS
4	OEM00356	ROCU10
5	ASSY800388-A	ROBOX OPERATOR BOX
6	BATTERY 2791 X2	MR-2791 Rechargeable Li-Ion Military Battery
7	ROC1-V01-0000#	RUGGEDIZED OPERATOR CONTROL UNIT
8	ASSY800160-A	ZIGBEE BOX
9	TOL00634	AIR PRESSURE GAUGE
10	ASSY800474-A	ROBOX COMMUNICATION BOX ASSY
11	ASSY800481-A	ROBOX TRIPOD RELAY CASE ASSY
12	TOL00968	DC/AC CURRENT VOLTAGE DIGITAL CLAMP MULTIMETER

4. Procedure:

Procedure			
Name			
Date			
Argo S/N			
Step	Description	P/F	Note
PERFORM A VISUAL INSPECTION – WITHOUT TRACKS			
1	**Ensure** there are no cracks and paint mismatches.		**Check** that the lower chassis and axles aren't deformed. **Check** that the vehicle's plastic cover is complete and not damaged.
2	**Verify** that all 6 top covers are secured with screws and washers.		
3	**Verify** that there are 5 bolts on each of the eight wheels.		**Verify** they are correctly locked.
4	**Check** for tire defects and ensure tire pressures are correct.		Wheel pressures vary with the vehicle's load. It is essential not to exceed 7 psi. **Front wheel** – 5 psi **2x Middle wheels** – 7 psi **Rear wheel** – 6 psi
5	**Verify** the existence of two emergency stop buttons on the platform with their covers (right and left).		**Verify** each button is secured to the frame with 6 screws.
6	**Verify** the existence and integrity of 4 peripheral cameras (Drive, Rear, Right, and Left)		**Verify** that the cameras are connected, the cables are in good condition, and there are no cracks/scratches on the lenses.
7	**Verify** the MAST camera's existence and integrity.		**Verify** that the camera is connected, the cable is in good condition, and there are no cracks/scratches on the lens.
8	**Verify** that 5 antennas are present and tightly connected.		4 RF antennas and 1 cellular&GPS.
9	**Perform** an engine oil level check and verify there are no oil leaks around the engine.		Oil level must be above the "L" mark and below the "F" mark on the dipstick.
10	**Perform** a brake fluid level check and verify there are no fluid leaks in the area.		

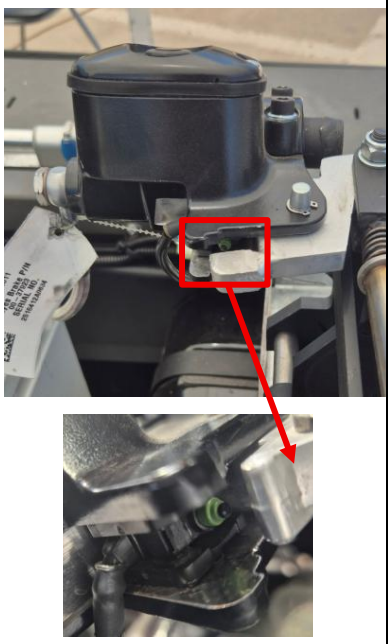
Step	Description	P/F	Note
11	Verify fuel level is sufficient for testing.		This can be verified by turning on the vehicle switch and seeing the fuel level on the dashboard.  Fuel level should not exceed the filler neck level. 
TURN THE SYSTEM ON (THE ENGINE IS OFF)			
1	Verify that the latest software version is up to date.		Make sure all software settings are ready.
2	Verify the NC switch is armed.		Verify that the software indicates the system is connected, and the Estop is off.
3	Check the 12V battery level on the front panel screen/ via the ROCU10.		The battery level must be at least 60% before starting testing.
4	Check the 48V battery level via the ROCU10.		The battery level must be at least 60% before starting testing.
5	Verify that there is a ping to all system components.		
6	Verify that the LEDS (front lights) are working.		
COMPONENTS CHECK			
1	Verify that all 5 days' cameras are streaming and in focus.		Drive, Rear, Right, Left, and Mast cameras.
2	Verify that the 4 thermal cameras are streaming and in focus.		Drive, Rear, Right, and Left cameras.
3	Verify the microphone is working, and the sound is clear.		Verify no delay
4	Verify that the speaker is working, and that the sound is clear.		Verify no delay
5	Verify that the software shows if the motors are off.		RPM = 0

GUN SYSTEM CHECK			
VISUAL INSPECTION			
Step	Description	P/F	Note
1	**Verify** that the weapon system is securely fastened with 4 screws and nuts, and that the 2 cables are correctly connected and undamaged.		
2	**Verify** that the weapon system cameras (day & thermal) are in good condition and that there are no cracks/scratches on the lenses.		
3	**Verify** the presence of a mechanical stopper on the elevation axis.		Ensure it locks manually. Leave it free on the axis after the check.
4	**Verify** that all components have no freedom.		
5	**Verify** that the charge status indicator on the control unit is lit.		
6	**Verify** that the blue LED at the bottom of the electronic drawer flashes.		Once the system is connected, it will stop flashing.
FUNCTIONAL INSPECTION			
1	**Verify** that the startup screen is displayed clearly.		
2	**Verify** the communication between the control unit and Pitbull is functioning correctly.		Ensure the communication icon on the control unit is green and that communication takes no more than a minute.
3	**Verify** that both mechanical stoppers on the axis are present, can be manually locked, and move freely along the axis.		Leave it free on the axis after the check.
4	**Verify** the weapon system cameras (day & thermal) are streaming and in focus.		
5	**Verify** the weapon system camera's zoom works.		
6	**Verify** the weapon system is moving in all directions.		Should move in X-Y-Z
7	**Perform** a full rotation of the system ($0° - 360°$) at slow speed.		Verify the freedom of movement.
8	**Verify** full elevation axis movement ($-30° - 60°$) at slow speed.		Verify the freedom of movement.
9	**Select** a target and verify that it remains at the center of the cross when zoomed.		
10	**Release** the safety mechanism and repeat the procedure with the firing reticle.		
11	**Select** a target and verify that it remains centered in the crosshairs with the thermal camera.		
12	**Release** the safety mechanism and repeat the procedure with the firing reticle.		
13	**Ensure** the gun system remains stable during motion.		

BEFORE TURNING ON THE ENGINE, MAKE SURE THE SHIFT IS SET TO – N MAKE SURE THE AREA IS CLEARED OF PEOPLE!			
START THE ENGINE AND DO AN EMERGENCY CHECK			
Step	Description	P/F	Note
	Perform an emergency check to ensure the 3 emergency stop buttons work – - Two on the platform. One in the operator box.		**Verify** that the engine turned off. **Verify** that the software shows if the Estop is on and the system is disconnected.
START THE ENGINE AND DO A DRIVING CHECK – WITHOUT TRACKS PLACE THE SYSTEM ON A LIFT FOR STEPS 1–4			
1	**Set** the system to charging mode and **measure** the alternator current; it should be above **40mV**.		Measure the current by placing TOL00968 on the alternator's red wire, connected to the 12 V battery. 
2	**Check** the brake actuator - • **Brake off**: Put the vehicle in N gear and press the throttle until the vehicle reaches 2,000 RPM. **Verify** the microswitch is pressed. • **Brake on**: Measure the piston length with the brake engaged and the microswitch not pressed.		 

Step	Description	P/F	Note
3	Check the RPM of the neutral gear shift modes: • Neutral standby (Engine off) • Neutral drive (Engine on)		In neutral standby, the RPM remains at 0 when the throttle is pressed. In neutral drive, the RPM changes when the throttle is pressed.
4	Driving and gear mode Check: Ensure that shifts can be changed from the controller by switching to manual mode. Verify that the system's right and left sides move smoothly and are synchronized. Turn right and left in low and high gear, and verify that the wheels move correctly.		Check R-N-D by pressing the throttle joystick for 20 sec. Perform the check in high and low modes. Right turn – the right wheels are locked at low mode, and only the left wheels move forward. Left turn – the left wheels are locked at low mode, and only the right wheels move forward.
5	Press the handbrake and ensure it engages and disengages properly. Verify that the software indicates whether the handbrake is disabled.		Check visually when the hood is open. Verify the piston is moving.
6	When shifting to R, ensure the display automatically switches from the drive camera to the rear camera.		
7	Drive the vehicle for 200 meters, in low and high gear, forward and backward, and verify that it moves in a straight line. Ensure the vehicle comes to a complete stop by applying the brakes.		Check when the vehicle is traveling at 15 km/h.
8	Drive the vehicle, turn right and left in low and high gear, and verify that the wheels move correctly.		Right turn – the right wheels are locked at low mode, and only the left wheels move forward. Left turn – the left wheels are locked at low mode, and only the right wheels move forward.
9	Perform a pivot in both directions and verify that the front wheel on that side remains stationary.		Make the turn at low gear.
10	Drive the vehicle for at least 2 hours on a flat surface in low and high gear, and verify that there is no loss of communication or any unusual noises.		Verify that the cameras are streaming, the drive is smooth, and communication is good.
11	Drive the vehicle uphill and downhill (max slope of 45°). Verify the brakes and ensure the vehicle remains stable and stationary without slipping.		Verify that there is no unusual noise.
12	Drive the vehicle on the side slope (max slope of 20°). Verify the brakes and ensure the vehicle remains stable and stationary without slipping.		Verify that there is no unusual noise.

Step	Description	P/F	Note
13	Perform an emergency check to ensure the emergency stop buttons work while driving.		Verify that the engine turned off. Verify that the software shows if the Estop is on and the system is disconnected.
PERFORM A VISUAL INSPECTION – WITH TRACKS			
1	Verify that the wheels have been switched to the opposite side for proper track installation.		This can be verified by measuring the distance from the edge of the vehicle's plastic to the wheel's end. With tracks - about 13 cm from the inside. 
2	Verify the existence of two mount tracks, their tension, and that their pins are secured.		
DRIVING CHECK – WITH TRACKS PLACE THE SYSTEM ON A LIFT FOR STEPS 1–4			
1	Set the system to charging mode and measure the alternator current; it should be above 40mV.		Measure the current by placing TOL00968 on the alternator's red wire, connected to the 12 V battery. 

Step	Description	P/F	Note
2	Check the brake actuator - • Brake off: Put the vehicle in N gear and press the throttle until the vehicle reaches 2,000 RPM. Verify the microswitch is pressed. • Brake on: Measure the piston length with the brake engaged and the microswitch not pressed.		
3	Check the neutral gear shift modes: • Neutral standby. • Neutral drive.		In neutral standby, the RPM remains at 0 when the throttle is pressed. In neutral drive, the RPM changes when the throttle is pressed.
4	Driving and gear mode Check: Ensure that shifts can be changed from the controller by switching to manual mode. Verify that the system's right and left sides move smoothly and are synchronized. Turn right and left in low and high gear, and verify that the wheels move correctly.		Check R-N-D by pressing the throttle joystick for 20 sec. Perform the check at high and low modes. Right turn – the right wheels are locked at low mode, and only the left wheels move forward. Left turn – the left wheels are locked at low mode, and only the right wheels move forward.
5	Press the handbrake and ensure it engages and disengages properly. Verify that the software indicates whether the handbrake is disabled.		Check visually when the hood is open. Verify the piston is moving.
6	When shifting to R, ensure the display automatically switches from the drive camera to the rear camera.		
7	Drive the vehicle for 200 meters, at low and high gear, forward and backward, and verify that it moves in a straight line. Ensure the vehicle comes to a complete stop by applying the brakes.		Check when the vehicle is traveling at 15 km/h.

Step	Description	P/F	Note
8	**Drive** the vehicle, turn right and left in low and high gear, and verify that the wheels move correctly.		**Right turn –** the right wheels are locked at low mode, and only the left wheels move forward. **Left turn –** the left wheels are locked at low mode, and only the right wheels move forward.
9	**Perform** a pivot in both directions and verify that the front wheel on that side remains stationary.		Make the turn at low gear.
10	**Drive** the vehicle uphill and downhill (max slope of 45°). **Verify** the brakes and ensure the vehicle remains stable and stationary without slipping.		**Verify** that there is no unusual noise.
11	**Drive** the vehicle on the side slope (max slope of 20°). **Verify** the brakes and ensure the vehicle remains stable and stationary without slipping.		**Verify** that there is no unusual noise.
12	**Drive** the vehicle on soft sand for **2 hours**. **Verify** the brakes and ensure the vehicle remains stable and stationary without slipping.		**Verify** that there is no unusual noise.
13	**Drive** the vehicle on deep sand, at max speed in low gear for 20 minutes.		Continuous drive must be maintained. Do not stop unnecessarily! **Verify** that there is no loss of communication or unusual noise.
14	**Drive** the vehicle on a rocky surface for **2 hours**. **Verify** the brakes and ensure the vehicle remains stable and stationary without slipping.		**Verify** that there is no unusual noise.
15	**Perform** an emergency check to ensure the emergency stop buttons work while driving.		**Verify** that the engine turned off. **Verify** that the software shows if the Estop is on and the system is disconnected.
COMMUNICATION RANGE CHECK			
1	**Verify** stable cellular communication.		**Verify** that the system detects all SIM cards.
2	**Verify** stable GPS communication.		**Verify** that the GPS parameters are changing.
3	**Verify** stable RF communication up to a range of 1 km, within line of sight.		**Check** RF communication without cellular. BLUSDR →PHASE 5

PERFORM A VISUAL INSPECTION BEFORE DELIVERY			
Step	Description	P/F	Note
1	**Set** the system to charging mode and **measure** the alternator current; it should be above **40mV**.		Measure the current by placing TOL00968 on the alternator's red wire, connected to the 12 V battery. 
2	Check the track tension.		
3	Check all component connections (mechanical and cables).		
4	Ensure there are no cracks and paint mismatches.		Check that the lower chassis and axles aren't deformed.
5	Verify the existence of 5 bolts on each of the eight wheels.		Verify they are correctly locked.
6	**Check** for tire defects and ensure tire pressures are correct.		Wheel pressures vary with the vehicle's load. It is essential not to exceed 7 psi. **Front wheel** – 5 psi **2x Middle wheels** – 7 psi **Rear wheel** – 6 psi
7	**Make** sure the antennas are connected and undamaged.		

Results Mentions Comments